

Complete Programming Course Structure

- Total Phase: 05
 - Total Lecture: 16
 - Total Problem Solving Day: 10
 - Total Assignment: 02
 - Exam: 01
-

Phase 01 — Fundamentals of Programming

Week 01 — Basic Programming Foundation

Lecture 01 — Basic Syntax, Variables & Data Types

- Introduction to Programming
- C Language Basics
- Input/Output
- Variables
- Data Types

Lecture 02 — Operators & Conditional Statements

- Arithmetic Operators
- Logical Operators
- Relational Operators
- if-else
- Nested Condition

Lecture 03 ---- Problem Solving Day 01

- Conditional Problem Solving
- Basic Logic Building

Lecture 04 — Loops

- for Loop
- while Loop
- do while Loop
- Nested Loop Basics

Lecture 05 --- Problem Solving Day 02

- Loop Problem Solving
- Number Problems
- Basic Patterns

Lecture 06 --- Assignment 01

- Condition + Loop Based Problems
-

Phase 02 — Logic Building & Arrays

Week 02 — Mastery in Logic & Introduction to Array

Lecture 07 — Advanced Conditional Problem Solving

- Complex Conditions
- Switch Case
- Logic Building

Lecture 08 — Advanced Problem Solving with Loops

- Nested Loops
- Digit Problems
- Series Problems

Lecture 09 — Problem Solving 03

- Pattern Problems
- Number Theory Basics

Lecture 10 — Introduction to Array

- Array Basics
- Input/Output
- Traversing Array

Lecture 11 — Problem Solving 04

- Array Manipulation
- Maximum/Minimum
- Reverse Array

Lecture 12 — Assignment 02

- Array + Loop Based Problems
-

Phase 03 — Arrays & String Manipulation

Week 03 — Array Operations & String Manipulation

Lecture 13 — Array Operations

- Insertion
- Deletion
- Searching
- Sorting Basics

Lecture 14 — Introduction to String

- Character Array
- String Input/Output
- ASCII Concept

Lecture 15 — Problem Solving 05

- String Problem Solving
- Palindrome
- Reverse String

Lecture 16 — String Operations & Functions

- strlen()
- strcpy()
- strcmp()
- strcat()

Lecture 17 — Problem Solving 06

- Advanced String Problems
- Character Frequency
- Password Validation

Phase 04 — Functions, Pointers & Nesting

Week 04 — Functions, Pointers & Nesting

Lecture 18 — Nested Loop & Pattern Printing

- Star Patterns
- Number Patterns
- Pyramid & Triangle

Lecture 19 — Functions in C/C++

- Function Declaration
- Parameters
- Return Type

Lecture 20 — Practice Day 07

- Function Based Problems

Lecture 21 — Pointers

- Memory Address
- Pointer Basics
- Pointer with Array

Lecture 22 — Problem Solving 08

- Pointer Arithmetic
- Swap using Pointer

Lecture 23 — Assignment 03

- Function + Pointer Problems

Phase 05 — 2D Array & Recursion

Week 05 — 2D Array and Recursion

Lecture 24 — Basics of Recursion

- Recursive Function
- Base Case
- Recursive Thinking

Lecture 25 — 2D Array (Matrices)

- Matrix Basics
- Matrix Operations
- Diagonal Problems

Lecture 26 — Problem Solving 09

- Matrix Problem Solving

Lecture 27 — Problem Solving with 2D Array & Recursion

- Recursive Problems
- Advanced Matrix Logic

Lecture 28 — Practice Day 10

- Mixed Problem Solving
- Mock Practice
- Debugging

Lecture 29 — Exam

- Fundamentals
- Loops
- Nested Loop
- Arrays
- 2D Arrays
- Strings
- Recursion
- Pointer